2025_Hexaware_Actual_Test_3

Test Summary

- No. of Sections: 2
- No. of Questions: 90
- Total Duration: 90 min

Section 1 - Aptitude

Section Summary

- No. of Questions: 60
- Duration: 60 min

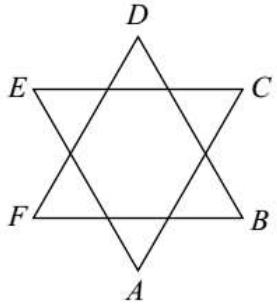
Additional Instructions:

None

Q1. In a class, the average mark obtained by a number of students in English is 52.25. 25% of students placed in C category made an average of 31 marks, while 20% who were placed in the A category made an average of 80 marks. Find the average marks of the remaining students.

- 50
- 52.2
- 51
- 51.8

Q2. In the adjoining figure ∠A + ∠B + ∠C + ∠D + ∠E + ∠F = ?



- 270°
- 300°
- 360°
- 330°

Q3. The value of $\log_{343}(7)$ is:

- $\frac{1}{3}$

$\frac{-1}{3}$

3

-3

Q4. A trapezium-based prism with two parallel sides 8 cm and 14 cm respectively and the distance between two parallel sides is 8 cm. Find the height of the prism if the volume of the prism is 1056 cm^3.

10 cm

12 cm

11 cm

8 cm

Q5. A hotel in Haridwar pays a servant Rs 90 plus one turban. The servant leaves the job after 9 months and receives Rs 65 and a turban. Then, find the price of the turban.

Rs 10

Rs 15

Rs 7.50

Cannot be determined

Q6. John has been given three different polynomial terms as follows:
$(x^2 - xy - 2y^2)$, $(2x^2 - 5xy + 2y^2)$ and $(2x^2 + xy - y^2)$

And he has been asked to find the Lowest Common Multiple for the given three polynomials. What will be its LCM?

$(x + y)(x^2 - 4y^2)$

$(x - y)(x^2 - 4y^2)$

$(x + y)(x + 2y)(2x - y)$

$(x + y)(x - 2y)(2x - y)$

Q7. Find the least number that when divided by 16, 18, and 20 leaves a remainder of 4 in each case, but is completely divisible by 7.

364

2254

2884

3234

Q8. In 1970, the cost of producing a litre of milk was Rs.11.25, 66% of which was the cost of food and the remainder was wages and other expenses. By 1973, the cost of food had increased by 55% and wages and other expenses had increased by 45%. Find the cost of producing a litre of milk in 1973.

Rs.17.055

Rs.19.678

Rs.12.293

None of these

Q9. A company could advertise its new product in 5 magazines, 4 newspapers, and 3 television channels. But in a later move, it decided to give advertisements in only 2 of the magazines, one of the newspapers and one of the TV channel. In how many ways can they advertise?

36

44

72

120

Q10. In how many ways 4 boys and 3 girls be seated in a row so that they are alternate?

144

288

12

256

Q11. The chances of failing in physics are 20% and the chances of failing in mathematics are 10%. What are the chances to fail in at least one subject?

28 %

18 %

15 %

10 %

Q12. If the selling price of 16 articles is equal to the cost price of 18 articles, then what is the profit percent?

12.5%

15%

18%

20%

Q13. An electronic weighing machine can show the correct weight within the range of 10 grams. Nathan used this machine and sells items at the cost price. If Nathan sells items only in the form of 400 grams packets, then what is the maximum loss he can suffer? (in %)

1.74

2.44

3.55

2.32

Q14. If the sixth term of an H.P. is 10 and the 11th term is 18, find the 16th term.

90

72

68

63

Q15. A can contains a mixture of two liquids A and B in proportion 7:5. When 9 liters of the mixture is drawn off and the can is filled with B, the proportion of A and B becomes 7:9. How many liters of liquid A was contained by the can initially?

18

21

24

32

Q16. From a full barrel containing 729 litres of honey, we pour off ‘a’ litre and add water to fill up the barrel. After stirring the solution thoroughly, we pour off ‘a’ litre of the solution and again add water to fill up the barrel. After the procedure is repeated 6 times, the solution in the barrel contains 64 litres of honey. Find a.

243 litres

81 litres

2.7 litres

None of these

Q17. A sum of Rs.5000 at CI becomes double in 5 years. In 20 years it will be

Rs.100000

Rs.80000

Rs.50000

Rs.20000

Q18. What annual payment will discharge a debt of Rs. 770 due after 5 years, the rate of interest being 5% per annum?

Rs 140

Rs 150

Rs 180

Rs 200

Q19. Bonar and Stanley can paint a masterpiece in 12 days and 60 days respectively. They were contracted to complete the masterpiece together for Rs. 24000. What will be the share of Stanley?

Rs. 18000

Rs. 1600

Rs. 20000

Rs. 4000

Q20. A train overtakes two men, walking in the same direction as the train, with speed of 3 km/hr and 4.5 km/hr in 8 and 12 seconds respectively. The length of the train is

10 m

20 m

30 m

40 m

Q21. Choose the appropriate article given in the options:

Copper is ____ useful metal.

a

an

the

zero article

Q22. In each question below is given a statement followed by two assumptions numbered I and II. You have to consider the statement and the following assumptions and decide which of the assumptions is implicit in the statement.

Should high chimneys be installed in industries?
Arguments:
1. Yes. It reduces pollution at ground level
2. No. It increases pollution in the upper atmosphere

Only 1 is true

Only 2 is true

Either 1 or 2 is true

Neither 1 nor 2 is true

Q23. **A number of sentences are given below which, when properly sequenced, form a coherent paragraph. Each sentence is labelled with a letter. Choose the most logical order of sentences from among the four given choices to construct a coherent paragraph.**

P) In 1995, astronomers at last found the first planet orbiting a faraway, sun-like star.
Q) Since then, however, astronomers have discovered thousands of exoplanets—or planets orbiting other stars—and many have conditions that could be favourable to life.
R) Six decades after Ozma, we know that planets vastly outnumber stars in the Milky Way, providing billions of places for alien metabolisms to emerge.
S) The massive Jupiter-like world, called 51 Pegasi b, is completely inhospitable for life as we know it.

PQSR

PSQR

SQRP

SRQP

Q24. Fill in the blank:
Listen to the story _____ answer the questions in complete sentences.

but

and

next

none of the above

Q25. Fill in the blank with a suitable adjective:

Every cloud has a _____ lining.

silver

first

foolish

none of the above

Q26. Read the sentence to find out whether there is any error in it. The number of that part is the answer. If there is no error mark 4 as the answer,

We forgave him (1) / because he (2) / confessed his crime (3) / No error (4)

1

2

3

4

Q27. **Spot the part that has an error in it.**

Each of the speakers whom I invited (A) / to participate in the debate has (B) / indicated their unwillingness (C) / because of the short notice (D).

- A
- B
- C
- D

Q28. In the following question, a sentence has been given in Direct/Indirect speech. Out of the four alternatives suggested, select the one that best expresses the same sentence in Indirect/Direct speech.

John said, "There is a monkey outside the window."

- John said that there was a monkey outside the window.
- John said that there is a monkey outside the window.
- John says that there was a monkey outside the window.
- John says that there is a monkey outside the window.

Q29. Identify the synonyms of the capitalized words.

FLAMBOYANT

- buoyant
- tarnish
- sluggish
- garish

Q30. **Choose the appropriate word that answers the description.**

One who abandons one's religion or party.

- altruist
- adherent
- apostate
- blowhard

Q31. Choose the word which conveys the **OPPOSITE** meaning for the word:

Profligate

- Cumbersome
- Lumpish
- Moral
- Prosaic

Q32. Change the voice in the given sentence.

You will not have eaten bread.

- Bread was not eaten by you
- Bread would not have eaten.
- Bread will not have been eaten by you
- Bread would have not eaten by you.

Q33. Select the related letters from the given alternatives.
G3S : J3P :: L4X : ?

- P3Y
- O3T
- O4T
- Q4S

Q34. Pointing to a boy, Ram said, 'He is the only son of my maternal grandfather's son'. How is the boy related to Ram?

- Son
- Brother
- Cousin
- Data Inadequate

Q35. On what dates of April 2001 did Wednesday fall?

- 2nd, 9th, 16th, 23rd
- 4th, 11th, 18th, 25th
- 12th, 18th, 27th, 6th
- 1st, 8th, 15th, 22nd

Q36. Find the mirror image of the clock when the time is 07:35.

- 4:25
- 5:25
- 4:30
- 6:25

Q37. In a certain code language, if the word SYMPHONY is coded as HBNKSLMB, then how is the word PICKLE coded in that language?

- KRXSNR
- KRWPLV
- KIXPOV
- KRXPOV

Q38. Which digit will appear on the face opposite to the face with number 1?





6

3

4

5

Q39. **The question below consists of a question and two statements numbered A and B given below it. You have to decide whether the data provided in the statements are sufficient to answer the questions. Read both statements and give an answer.**

Suresh gets two successive discounts on purchasing a tape recorder. Find out the second discount.
(A) Selling Price and Marked Price of the tape recorder are Rs 300 and Rs 400 respectively.
(B) The first discount is 20% less than the second discount percentage.

if statement (A) alone is sufficient to solve the question, but statement (B) alone is not

if statement (B) alone is sufficient to solve the question, but statement (A) alone is not

if neither statement (A) nor statement (B) is individually sufficient to solve the question, but a combination of both is sufficient to solve the question

if both the statements(A) and (B) are individually sufficient to solve the question

if both the statements taken together are not sufficient and more information is required to solve the question

Q40. The question below consists of a question and two statements numbered A and B given below it. You have to decide whether the data provided in the statement(s) is/are sufficient to answer the question. Read both statements and give an answer.

If X, Y and Z are positive, is $X=\frac{Y}{Z^{2}}$?
(A)$Z=\sqrt{\frac{Y}{X}}$
(B)$Z=\frac{Y}{X\times Z}$

if statement (A) alone is sufficient to solve the question, but statement (B) alone is not.

if statement (B) alone is sufficient to solve the question, but statement (A) alone is not.

if neither statement (A) nor statement (B) is individually sufficient to solve the question, but a combination of both is sufficient to solve the question.

if both the statements(A) and (B) are individually sufficient to solve the question.

Q41. Cavin walked 100m towards the east took a right turn and walked 400 m. Then he took a left turn and walked 300m.In which direction is he now from the starting point?

- Southwest
- Southeast
- Northeast
- Northwest

Q42. Find the missing term in the following number series.

28, 29, ?, 195, 796, 4005

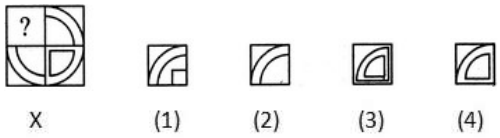
- 62
- 68
- 81
- 83

Q43. In the question below are given two or three statements followed by two conclusions numbered I and II. You have to take the given statements to be true even if they seem to be at variance from commonly known facts and then decide which of the given conclusions logically follows from the statements, disregarding commonly known facts. Give answer.

Statements:
Only a few Cow is Goat.
No Kite is Sheep.
Few Cow is Sheep.
Conclusions:
I. All Cow being Goat is a possibility.
II. Some Kite is not Goat.

- Both conclusion I and conclusion II follow
- Only conclusion II follows
- Only conclusion I follows
- Neither conclusion I nor conclusion II follows

Q44. Identify the figure that completes the pattern.



1

2

3

4

Common Content:

Indian cities offer new _____1_____ for development across various urban sectors, but most of these cities ____2_____ from urban flooding during monsoon. This eventually leads to loss of life and property and ____3_____the economic activities and natural eco-system.Even the most planned Indian cities suffer from this problem. For instance, when Chandigarh, which is ___4_____ known as the first planned city of independent India, receives high rainfall, _____5_____ find it difficult to reach their destinations.

Q45. Choose the right word for the blank numbered (4)

angrily

perfectly

famously

proudly

Q46. Choose the right word for the blank numbered (1)

adversity

opportunities

threat

dedication

Q47. Choose the right word for the blank numbered (5)

transporters

commuters

canard

guise

Q48. Choose the right word for the blank numbered (3)

- affects
- effects
- agitate
- consequences

Q49. Choose the right word for the blank numbered (2)

- suffer
- agonize
- endure
- tolerate

Common Content:

Read the passage and answer the following questions:

The Stratosphere, specifically, the lower Stratosphere has, it seems, been drying out. Water vapor is a greenhouse gas, and the cooling effect on the Earth's climate due to this desiccation may account for a fair bit of the slowdown in the rise of global temperatures seen over the past ten years. The Stratosphere sits on top of the Troposphere, the lowest, densest layer of the atmosphere. The boundary between the two, the Tropopause, is about 18km above your head if you are in the tropics, and a few kilometers lower if you are at higher latitudes (or up a mountain). In the Troposphere, the air at higher altitudes is in general cooler than the air below it, an unstable situation in which warm and often moist air below is endlessly buoying up into cooler air above. The resultant commotion creates clouds, storms, and much of the rest of the world's weather.

In the Stratosphere, the air gets warmer at higher altitudes, which provides stability. The Stratosphere-which extends up to about 55km, where the Mesosphere begins, is made even less weather-prone by the absence of water vapor, and thus of the clouds and precipitation to which it leads. This is because the top of the Troposphere is normally very cold, causing ascending water vapor to freeze into ice crystals that drift and fall, rather than continue up into the Stratosphere. A little water manages to get past this cold trap. But as Dr. Solomon and her colleagues note, satellite measurements show that rather less has been doing so over the past ten years than was the case previously. Plugging the changes in water vapor into a climate model that looks at the way different substances absorb and emit infrared radiation, they conclude that between 2000 and 2009 a drop in the Stratospheric water vapor of less than one part per million slowed the rate of warming at the Earth's surface by about 25%. Such a small change in Stratospheric water vapour can have such a large effect precisely because the Stratosphere is already dry. It is the relative change in the amount of greenhouse gas, not its absolute level, which determines how much warming it can produce.

Q50. What is the order of layers in the atmosphere, starting from the lowermost and going to the topmost?

- Tropopause, Troposphere, Mesosphere, Stratosphere
- Troposphere, Tropopause, Stratosphere Mesosphere
- Troposphere, Tropopause, Mesosphere, Stratosphere
- Troposphere, Stratosphere, Tropopause, Mesosphere

Q51. What in the passage has been cited as the main reason affecting global temperatures?

Relative change in water vapour content in the Stratosphere

Drop in Stratospheric water vapour of less than one part per million

The extreme dryness in the Stratosphere

Absorption and emission of infrared radiation by different substances

Q52. Why is the situation in the troposphere defined as unstable?

Because, unlike the Stratosphere, there is too much water vapour in the Troposphere

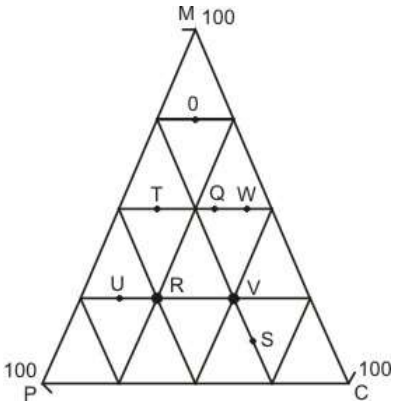
Because the Troposphere is not directly linked to the Stratosphere, but through the Tropopause which creates much of the world's weather

Because of the interaction between warm and cool air which is unpredictable in nature and can lead to storms

Because this layer of the atmosphere is very cloudy and can lead to weather-related disruptions

Common Content:

The following triangular bar diagram represents the percentages of students offering Physics, Chemistry, and Mathematics in various colleges. Refer to the diagram to answer the questions that follow.



Q53. The total number of students in colleges R and V are 600 and 1000 respectively. How many students are studying Physics at R and V?

500

600

450

550

Q54. If the total number of students in colleges O, Q, R, and S are 600, 750, 900, and 1200 respectively, which college had the largest number of students taking chemistry?

- O
- Q
- R
- S

Q55. Among the colleges U, R, V and S, the college(s) where less than 50% of the students took Maths, is/are

- S
- V
- U and R
- U, R, V, S

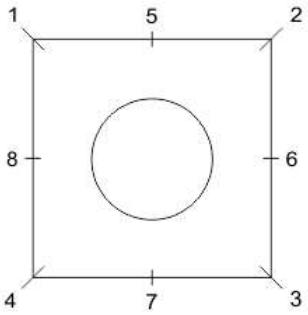
Q56. Colleges O and T have 600 students each. How many of these are studying Mathematics?

- 650
- 750
- 950
- 800

Common Content:

Study the following information carefully and answer the questions given beside:

Eight persons – Dinesh, Arush, Jolly, Gopal, Fahim, Govind, Hitesh and Jacky are sitting around the inner circle at an equal distance from each other. Four of them are facing towards the center while four are facing outside the center.



Only two persons are sitting between Arush and Dinesh. Hitesh sits second right of Dinesh. Gopal and Hitesh are immediate neighbours. Jolly sits third left of Gopal. Both Dinesh and Arush are facing outside the center. Gopal is facing the opposite direction of Arush. Fahim sits immediate right of Hitesh who is facing the same direction as Dinesh. Jacky sits second left of Jolly. Both Jacky and Fahim are facing the same direction.

Now they start playing cards game. They shuffled a pack of cards. Each of them draws a card according to the alphabetical order of their names and changes position according to the conditions given below:

1. If the card drawn is a spade, the person who draws first moves to position 5 facing the opposite direction of the current direction, then the second person who also draws a spade moves to position 6, similarly 7 and 8.
2. If the card drawn is a heart, the person remains in the same position facing the same direction.
3. If the card drawn is a diamond, the person who draws first moves to corner 1 facing the same direction, then the second person, who also draws diamond moves to corner 2, similarly 3 and 4.
4. If the card drawn is a club, the person remains in the same position facing the opposite direction of the current direction.

Draws:

A. Dinesh draws Ace of Club.
B. Arush draws 8 of Club.
C. Jolly draws 10 of Spade.
D. Gopal draws a King of Diamond.
E. Fahim draws Jack of Spade.
F. Govind draws Queen of Heart.
G. Hitesh draws 2 of Heart.
H. Jacky draws 7 of Diamond.

Q57. After all, persons changed their places based on the above conditions, then who among the following was not sitting in the circle?

- Govind
- Arush
- Hitesh
- Other than these options

Q58. After all persons changed their places based on the above conditions, then who among the following acquired the lowest position in the rectangle?

- Gopal
- Ayush
- Fahim
- Jacky

Q59. After all persons changed their places based on the above conditions, then both position and direction of how many person(s) is the same as before?

- None
- Two
- More than three
- One

Q60. After all persons changed their places based on the above conditions, then who among the following was sitting at the position 6 of the rectangle?

Dinesh

Govind

Jolly

Jacky

Section 2 - Pseudocode and Computer Fundamentals

Section Summary

- No. of Questions: 30
- Duration: 30 min

Additional Instructions:

None

Q1. What will be the output of the following pseudocode for a= 3 and b= 4?

```
1 integer foo( integer a, integer b)
2 if( a EQUALS 0)
3 return b
4 else if ( a EQUALS 1)
5 return b + 1
6 else
7 return foo(a - 1, a + b +5)
8 end if
9 end function foo()
```

20

10

7

5

Q2. What will be the output of the following pseudocode for a= 2?

```
1 integer fun( integer a)
2 integer p=1
3 if( a > 0)
4 return p + fun( a- 1)
5 else
6 p= p + 1
7 return a + p
8 end if
9 end function fun()
```

6

4

8

2

Q3. What will be the output of the following pseudocode?

```
integer a[10]
set a[10]= { 1,2,3,4,5,6,7,8,9}
for(each i from 1 to 4)
if (a[ i / 2])
print a[ i * 2]
else
print a[ i + 2]
end if
end for
```

2 4 6 8

1 2 3 4 5

2 6 8 0

3 5 7 9

Q4. What will be the output of the following pseudocode?

[Note: The bitwise exclusive OR operator(^) compares each bit of its first operands to the corresponding bit of its second operator. The corresponding result bit is set to 1. Otherwise, the corresponding result bit is set to 0].

```
integer a,b
set a= 0
a= a^ 1
a= a^ 2
a= a^ 3
print a
```

3

0

1

2

Q5. What will be the output of the following pseudocode?

```
integer h
set h= 42
do
print h
h ++
if (h > 45)
jump out of the loop
while()
end do while
```

42 43 44 45

43

42

42 44 45 46

Q6. What will be the following pseudocode?

[Not:&& : Logical AND - The logical AND operator (&&) returns the boolean value True (or 1) if either or both operands is True and returns False(or 0)].

```
1  character p[20], b[20]
2  set p[20]= "Respect Your elders"
3  integer k
4  set k= 0
5  while (p[k] not equals null character AND p[k] not equals space)
6  b[k]= p[k]
7  k= k + 1
8  end while
9  set k =0
10 while( b[k] not equals null character)
11 print b[k]
12 k= k + 1
13 end while
```

Respect elders

R

elders

Respect

Q7. What will be the output of the following pseudocode?

```
1  Set k = 1, m = 9
2
3  If (k + 8 / m) {
4     Print m
5  }
6
7  Set m = m + 1
8
9  If (k * 8 / m) {
10    Print k
11 }
12
13 Set k = k - 1
14
15 For each k from m down to 8 {
16    Print k
17 }
```

9 10 9 9

9 10

9 10 9 8

9 10 9 10

Q8. What will be the output of the following pseudocode?

```
1  integer s
2  set s=78
3  while(s EQUALS 78)
4  if ( s< 89)
5  s= s + 1
6  print "welcome"
7  else
8  jump out of the loop
9  end if
10 end while
```

None of the mentioned options

It will print "welcome" only one time

It will print nothing

It will print "welcome" 78 times

Q9. What will be the output of the following pseudocode for a=3?

```
1  integer fun(integer a)
2  integer arr[]={10,7,2,6,1,2,3}
3  integer p, sum =0
4  for(each p from 0 to a-1)
5  sum = sum + arr [a]
6  end for
7  return sum
8  end function fun()
```

20

19

21

15

Q10. What will be the output of the following pseudocode?

```
1  integer arr[]= {10, 20,30,40,5}
2  integer a, s
3  set s=0
4  set a= arr[ 1] +arr [2]
5  print a
```

50

40

25

5

Q11. What will be the output of the following pseudocode?

In Python ^ is used to find the power of a number.

```
1 integer a, b,c
2 set b=3, a= 5, c=1
3 if( b < a )
4 b= a
5 end if
6 for(each b from 0 to (1^2^3))
7 b= b + 1
8 end for
9 print b
```

6

5

2

1

Q12. What will be the output of the following pseudocode?

```
1 integer a, b
2 set a=15, b= 7
3 a= a mod (a-3)
4 b= b mod (b-3)
5 a= a mod 1
6 b= b mod 1
7 print a + b
```

0

7

15

2

Q13. What will be the output of the following pseudocode for a=2?

```
1 integer fun(integer a)
2 if(a>0)
3 return a+ fun(a-1)
4 else
5 return 0
```

```
6  end if
```

- 6
- 5
- 3
- 4

Q14. What will be the output of the following pseudocode?

```
1  integer a, b,c
2  set b=5, a=2, c=2
3  if(a)
4  a=a-1
5  a=c
6  else
7  c=c+1
8  end if
9  if(a-1)
10 c=c+1
11 else
12 c=c-1
13 end if
14 print a+b+c
```

- 15
- 20
- 10
- 11

Q15. What will be the output of the following pseudocode f(x) x =777.

```
1  Integer fun(Integer x)
2        if (x > 12)
3          fun (x / 9)
4          print x
5        else
6          for (each i from 1 to 3)
7              x = x + i
8              print x
9          end for
10       end if
11 end Function fun( )
```

- 11 13 14 85 777
- 12 15 86 777
- 10 12 15 86 777

10 12 13 85 777

Q16. Which of the following can be done if the application we are working on gets frozen?

Wait till it gets over

Open windows task manager and force closing the application

None of the mentioned options

Shut down the computer

Q17. Someone tells you that he is receiving a lot of mail from sources he does not know about products & services that he is not interested in. What is he likely to be getting?

Viruses embedded in emails

Mails from partners of shops he has purchased products from

Personalized marketing emails from product suppliers

None of the mentioned options

Q18. Magnetic tape is an example of which of the following storage devices?ru

Secondary storage devices

Tertiary storage devices

Primary storage devices

None of the mentioned options

Q19. Which of the following technologies enables us to make voice calls over the network?

Internet voice protocol

Digital telephony subscriber service

Internet protocol

Voice over internet protocol (VoIP)

Q20. In which of the following cases can we use “Forward” feature in email?

1. To send the email to a colleague who is not on the To or Cc lines.
2. To reply to the original sender and all other recipients on the To and Cc.

Choose the correct answer from the options given below.

- Both 1 and 2
- Only 2
- Neither 1 nor 2
- Only 1

Q21. After inserting an image in a Word document, which of the following keys should be used to resize the image by dragging while maintaining the image's proportions?

- Tab
- Alt
- Ctrl
- Shift

Q22. Which of the following is a file on a computer that links to another program or file somewhere else on the same computer?

- Shortcut
- File link
- Attachment
- URL

Q23. In DBMS, there are two tables, Teacher (containing information about all the teachers in a school) and Class (containing information about classes taught by the teachers). Which of the following relationships will link them?

- One-to-one
- None of the mentioned options
- Many-to-many
- One-to-many

Q24. The ALU of a computer normally contains which of the following high-speed storage elements?

- Hard disks
- Magnetic disk
- Semiconductor memory
- Registers

Q25. Which of the following cannot be part of an email address?

- Characters
- Dot
- Space
- Number

Q26. Which of the following programs automatically connects to websites and download documents and save them to the local drive?

- Offline browser
- Web servers
- Web downloading utilities
- None of the mentioned options

Q27. In the railway reservation system, the entities are date, PNR number, place of departure, destination, type of birth and seats available. Which of the following is the primary key?

- PNR number and place of departure
- PNR number
- PNR number and date
- PNR number and destination

Q28. Which of the following is a technique that marked the beginning of computer communications?

- User Environment

Batch Environment

Time Sharing

Message passing

Q29. Which of the following functions in MS Excel is used for error handling?

1. IF
2. IFS
3. IFNA

Only 3

Only 1

1, 2 & 3

Only 2

Q30. Which of them is an example of grabbing email information?

Cookie stealing

Reverse engineering

Port scanning

Banner grabbing

Answer Key & Solution

Section 1 - Aptitude

Q1

51.8

Solution

Let there are 100 students in the class, then in category A = 20 students, in C = 25 students, remaining = 55 students. Let x be the average of these 55 students.

So 20*80+ 25*31 + 55*x = 52.25*100 = 51.8

Q2

360°

Solution

In ΔACE

∠A + ∠C + ∠E = 180°

Similarly in ΔDFB

∠D + ∠F + ∠B = 180°

∴ (∠A + ∠C + ∠E) + (∠D + ∠F + ∠B) = 360°

∠A + ∠B + ∠C + ∠D + ∠E + ∠F = 360°

Q3

$\frac{1}{3}$

Solution

$\log_{343}7$ = $\log_{7^{3}}{7}$

= $1/3 \log_7 7$

= 1/3

Q4

12 cm

Solution

Volume of the prism = 1056 cm^3

WKT, **Volume of prism = Base x Height**

Base area = Area of trapezium

Area of trapezium = (1/2)(a + b)*h = (1/2)[8 + 14]*8

= (1/2)*22* 8 = 88 cm^2

Volume of prism = (Area of trapezium) x Height

1056 = 88 * height

Height of prism = 12 cm.

Q5

Rs 10

Solution

Given, (90 + T) * 9/12 = 65 +T

Or 3/4 (90 +T)

= (65 + T)260 + 4T = 270 + 3T

=> (4T – 3T) = (270 -260)

Or T = Rs 10

Q6

(x + y) (x – 2y) (2x – y)

Solution

The factrors of x^2-xy-$2y^2$ are

x^2-xy-$2y^2$

=x^2-2xy+xy-$2y^2$

=x(x-2y)+y(x-2y)

=(x-2y)(x+y)

The factors of $2x^2$-5xy+$2y^2$

=$2x^2$-4xy-xy+$2y^2$

=2x(x-2y)-y(x-2y)

=(x-2y)(2x-y)

The factors or $2x^2$+xy-y^2

=$2x^2$+2xy-xy-y^2

=2x(x+y)-y(x+y)

=(x+y)(2x-y)

Therefore the L.C.M is (2x-y)(x-2y)(x+y)

Q7

2884

Solution

All the four options are divisible by 7.

The number should leave the remainder when divided by 16, 18, 20.

LCM of 16, 18, 20 is 720.

Only the option (2884) leaves remainders 4 when divided by 720.

Q8

Rs.17.055

Solution

In 1970, cost of producing a litre of milk = Rs.11.25. Cost of food = 66% = Rs. 7.425.

∴ Wages and other expenses = Rs. 3.825.

In 1973, cost of food = (1.55) (7.425) = Rs. 11.50.

In 1973, wages and other expenses = 1.45 x.3.825 = Rs. 5.55

∴ In 1973, cost of producing a litre of milk =11.50 + 5.55 = Rs.17.05 (approx.).

Q9

120

Solution

Choosing 2 magazines out of 5 magazines in $5C_2$ ways = 10 ways and choosing 1 newspaper out of 4 newspaper in $4C_1$ ways = 4 ways

and choosing 1 T.V out of 3 T.V in $3C_1$ ways = 3 ways

=> 10 * 4 * 3 = 120

Total 120 ways.

Q10

144

Solution

Let the Arrangement be,

B G B G B G B

4 boys can be seated in 4! Ways

Girl can be seated in 3! Ways

Required number of ways,

= 4! × 3!

= 144

Q11

28 %

Solution

Let P(A) = 20/100 = 1/5,

P(B) = 10/100 = 1/10

Since events are independent and we have to find

P(A U B) = P(A) + P(B) – P(A). P(B)

= 1/5 +1/10 – 1/5 × 1/10

= 3/10 – 1/50

= 14/50

= 14/50 × 100 = 28%

Q12

12.5%

Solution

The selling price of 16 Articles = Cost price of 18 Articles

Let, the Cost Price of 1 Article = C.P

S.P of 1 Article = 18 C.P/16

Required profit percent = (18C.P/16 - C.P)/C.P × 100 = 12.5%.

Q13

2.44

Solution

Nathan will suffer maximum loss when Machine shows 410 grams as 400 grams

He will get price for 400 grams but will give 410 grams

∴ Maximum Loss percentage = (410 - 400)/410 × 100 = 2.44%

Q14

90

Solution

a + 5d = 1/10

a + 10d = 1/18

Solving these 2 equations, we get

a = 13/90 and d = (-2/225)

Hence, 16th term = a + 15d = (13/90) - 15(2/225) = 1/90

Q15

21

Solution

Let the can initially contain 7x litre and 5x litre of mixture A and B respectively.

Thus, out of 12x litre of the total mixture, 9 litres were taken out.

Quantity of A in mix. left= 7x - 9(7/12) = (28x - 21)/4 litre

Quantity of B in mix. left= 5x- 9(5/12) + 9 = (20x+21) / 4 litre

Resultant ratio = 7:9

[(28x - 21)/4] / [(20x+21) / 4] = 7/9

x = 3

Initially, liquid A in the can = 7x = 21 litres

Q16

243 litres

Solution

Initial honey =729

at last =64

After removal and replacement,

Final Concentration = Initial Concentration [1 - (R/Tq)] ^{n}

R - Quantity removed and replaced; TQ = Total quantity; n = Number of times replaced

64 = 729(1 - a/729)^{6}

64/729 =(729 – a/729)^{6}

(2/3) =(729 –a)/729

729 –a =486

a = 243

Q17

Rs.80000

Solution

In 5 years.....Rs.10000, in 10 years.....Rs.20000, in 15 years.... Rs.40000, in 20 years ..... Rs.80000.

Q18

Rs 140

Solution

Let the annual payment be Rs. P

the amount of Rs p in 4 years at 5% =100P + 4 × 5P/100 =120 P/100

The amount of RS p in 3 years at 5% =115P/100

The amount of Rs P in 2 years at 5% =110P/100

The amount of Rs P In 1 years at 5% =105P/100

These four amounts together with the last annual payment of Rs P will discharge the debt of Rs 770

120P/100 + 115P/100 + 110P/100 + 105P/100 + P =770

550 P/100 =770

P =770 × 100/550 =140

Hence annual payment =Rs 140

Q19

Rs. 4000

Solution

Total work= 60 units

Efficiency of Bonar = 60/12 = 5 units/day

Efficiency of Stanley = 60/60 = 1 unit/day

Salary Ratio of Bonar and Stanley = 5:1

Stanley's Salary= (1/6)* 24000= Rs. 4000

Q20

10 m

Solution

Let the length of the train be X m and its speed be S m/sec.

=> Time required by train = Distance/ Relative speed

=> Speeds of the two men are 3 x 5/18 = 5/6 m/s and 4.5 x 5/18 = 5/4m/s.

=> 8 = X/(S – 5/6) and 12 = X / (S – 5/4)

Divide the two equations to get the value of S = 2.09m/sec.

=> Length of train = X = 10.08m. Ans. (1)

Q21

a

Solution

Articles are words that define a noun as specific or unspecific.

The articles 'a' and 'an' are used with the name of things that are not specific.

The article 'an' is used with the words that begin with the sound of a vowel.

Any abbreviation having the sound of a vowel will be preceded by the article 'an'.

For example:

He is a MLA. (Wrong)

He is an MLA. (Correct)

In the given sentence, the word 'useful' starts with a vowel but has the sound of a consonant.

Hence, the article 'a' will be the correct choice here.

Q22

Only 1 is true

Solution

Pollution at ground level is the most hazardous in the way of being injurious to human and animal life. So argument 1 alone holds

Q23

PSQR

Solution

Solution:

Here the statement tells us about the discovery of new planets. Sentence P tells first time scientists find a planet orbiting a star. Followed by S that tells about the planet.

Sentence Q tells after that scientists find many planets. So the correct sequence would be PSQR.

Q24

and

Solution

and-used to connect two clauses when the second refers to something that happens after the first.

Q25

silver

Solution

Silver; Although Silver and Foolish are both descriptive adjectives, Silver fits the best here

Q26

4

Solution

There is no error in the given statement, as there are no grammatical errors in it.

Q27

C

Solution

'Indicated their unwillingness' in the sentence should be replaced by 'indicated his unwillingness'.

In referring to any body, everybody, everyone, anyone, each etc., the pronoun of the masculine or the feminine gender is used according to the context. But when the gender is not determined, we use the pronoun of the masculine gender, as there is no singular pronoun of the third person to represent both male and female.

Q28

John said that there was a monkey outside the window.

Solution

Here the reporting verb is in Simple Past, so the Simple Present tense in the reported speech will be changed to the corresponding Simple Past tense in Indirect form. Thus, ‘is’ will be replaced by ‘was’.

Q29

garish

Solution

Flamboyant means tending to attract attention because of their exuberance, confidence, and stylishness.

buoyant means cheerful, floating, tarnish means to stain, sluggish means “slow”, garish means “too showy”

Q30

apostate

Solution

An apostate is a person who renounces a religious or political belief or principle.

Q31

Moral

Solution

Explanation:

Profligate means someone who behaves in a way that is

morally wrong or who spends and wastes a lot of money.

Cumbersome - large or heavy and therefore difficult to

carry or use; unwieldy.

Lumpish - roughly or clumsily formed or shaped.

Moral - concerned with the principles of right and wrong

behaviour.

Prosaic - commonplace; unromantic.

Q32

Bread will not have been eaten by you

Solution

The given sentence is in the future perfect tense.

The Passive verb form: will/shall + have been + past participle form of the verb.

The appropriate choice would be "Bread will not have been eaten by you"

Q33 O4T

Solution

G is coded as J (+3)

S is coded as P (-3)

Similarly L4X is coded as O4T

Q34 Cousin

Solution

Ram's maternal grandfather is his mother's father.

The son mentioned must be Ram's uncle.

So, the boy is the only son of Ram's uncle.

This means the boy is Ram's cousin.

Q35 4th, 11th, 18th, 25th

Solution

We shall find the day on 1st April 2001.

1st April, 2001 = (2000 years + Period from 1.1.2001 to 1.4.2001)

Odd days in 1600 years = 0

Odd days in 400 years = 0

Jan. Feb. March April

(31 + 28 + 31 + 1) = 91 days 0 odd days.

Total number of odd days = (0 + 0 + 0) = 0

On 1st April 2001, it was Sunday.

In April 2001 Wednesday falls on the 4th, 11th, 18th, and 25th.

Q36 4:25

Solution

Mirror image= 11:60 - 07:35= 04:25

Q37 KRXPOV

Solution

Word: S Y M P H O N Y

Pattern: To each of these letters their opposite letters from the other end of the alphabet is given.

Code: H B N K S L M B similarly, the code for PICKLE is KRXPOV

Q38 6

Solution

When we fit Figure 1 in Figure 2, then we able to find 6 is opposite of 1.

Q39 if neither statement (A) nor statement (B) is individually sufficient to solve the question, but a combination of both is sufficient to solve the question

Solution

From Statement (A), we get the Selling Price and Marked Price which are not sufficient.

From Statement (B): Let the second discount percent = x, then the first discount percent =80% of x = 0.8x. This alone is not sufficient.

Combining both, we get: 400 = 300 ((100-0.8x)/100) ((100-x)/100)

Clearly, x can be found out

Q40 if both the statements(A) and (B) are individually sufficient to solve the question.

Solution

Statement (A): Since X, Y and Z are positive, both sides of the equation can be multiplied by X/Z to produce the identical equation $X = Y/Z^2$.

Statement (B) is also an equivalent equation, which can be seen by squaring both sides of the equation and then solving for X. Hence each statement gives sufficient information.

Q41 Southeast

Solution

No Solution

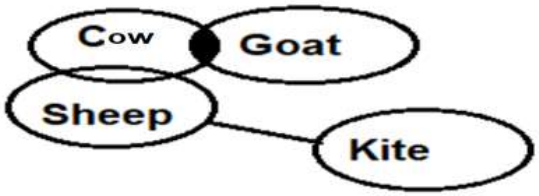
Q42 62

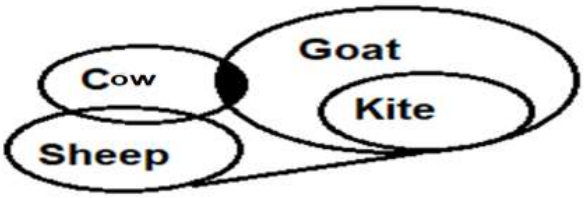
Solution

No Solution

Q43 Neither conclusion I nor conclusion II follows

Solution





Since only a few Cow is Goat is given which means some Cow is definitely not Goat so, the conclusion I does not follow.

All Kite can be Goat

So, the conclusion II also does not follow.

Q44 4

Solution



Q45. famously

Solution

first planned city of India a positive comment and the word famous is the right option

Q46. opportunities

Solution

The first blank should clearly have a 'positive' word .

Also the first blank should have a noun.

Hence, Opportunities is the correct answer

Q47. commuters

Solution

No Solution

Q48. affects

Solution

loss of life and property eventually affects the economy

Q49. suffer

Solution

No Solution

Q50. Troposphere, Tropopause, Stratosphere Mesosphere

Solution

From the passage, the order of layers is Troposphere, Tropopause, Stratosphere Mesosphere

Q51. Drop in Stratospheric water vapour of less than one part per million

Solution

From the last few lines of the passage. The stratospheric region is the atmosphere above 50 km above Earth's surface. The stratospheric water vapors add to the global warming in the atmosphere.

Q52. Because of the interaction between warm and cool air which is unpredictable in nature and can lead to storms

Solution

The situation in the troposphere is defined as unstable because the reaction of the natural atmospheric phenomenon like climate, temperature, humidity, rainfall, etc. are not realists predictable and stable.

Q53. 550

Solution

In R, 50% are studying Physics and in V, 25% are studying physics.

Hence,50/100 * 600 + 25/100 * 1000 = 550

Q54. S

lution

College	% offering Chemistry	No. of student offering chemistry
O	12.5	75
Q	33.33	250
R	25	225
S	62.5	750

his can be inferred from the diagram since S among O, Q, R, and S is larger to C.

Q55. U, R, V, S

Solution

All colleges below like TQW will have less than 50% of students offering Maths.

Q56. 750

Solution

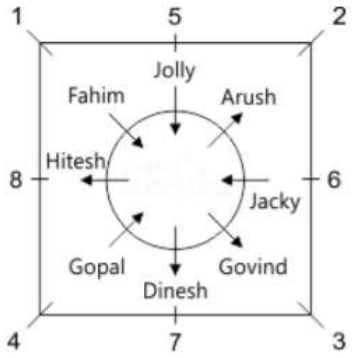
In college O, 75% are studying Maths=>75/100*600=450

In college T,50% are studying Maths=>50/100*600=300

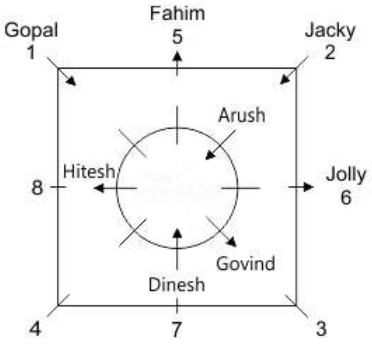
Hence in colleges O and T, i.e.450+300=750 students are studying Maths.

Q57. Other than these options

Solution

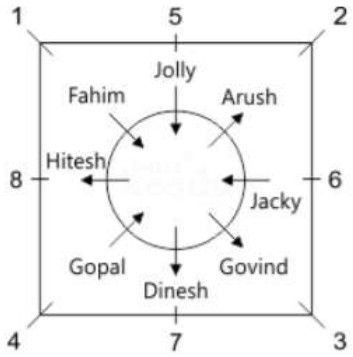


After draws, new location will be

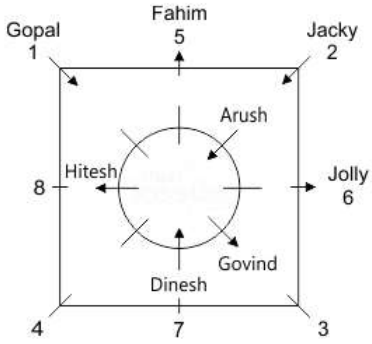


Q58. Gopal

Solution



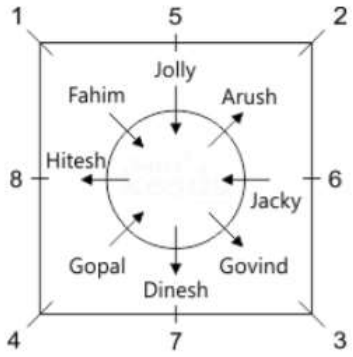
After draws, new location will be



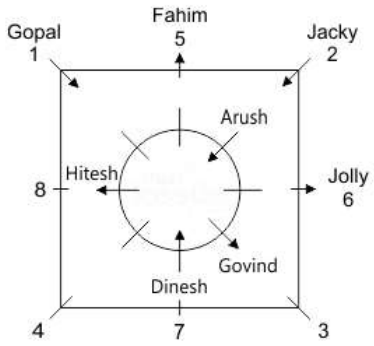
Gopal acquired the lowest position in the rectangle.

Q59. Two

Solution



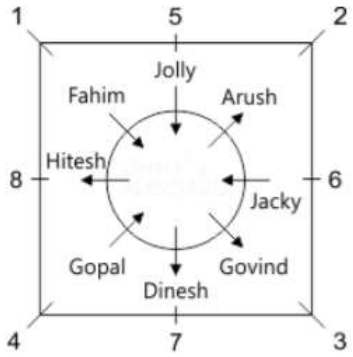
After draws, new location will be



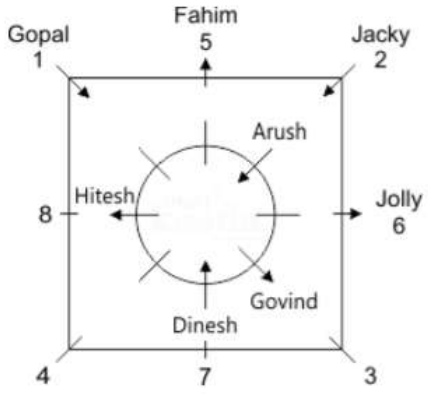
wo persons are same as before.

Q60. Jolly

Solution



After draws, new location will be



Jolly was sitting at the position 6 of the rectangle.

Section 2 - Pseudocode and Computer Fundamentals

Q1

20

Solution

foo(3,4) --> foo(2,12) --> foo(1,19) [returns 19+1 = 20]

20 value returned to the first function call and gets printed.

Q2

4

Solution

fun(2) --> return 1+fun(1) --> return 1+fun(0)

In fun(0) , p=1+1 = 2 . returns 0+2 = 2

In fun(1) , 1+ 2 = > returns 3

In fun(2), 1+3 => returns 4

Q3

3 5 7 9

Solution

Array values a[10]= { 1,2,3,4,5,6,7,8,9}.

i=1=> a[1/2] => a[0] = 1 => if block prints a[1*2] = **3**

i=2=> a[2/2] => a[1] = 2 => if block prints a[2*2] = 5. Similarly loop executes for i=3,4 and prints 7 and 9.

Q4

0

Solution

0=> 0000

1=> 0001

2=> 0010

3=> 001 1

---

^=> 0000

Q5

42 43 44 45

Solution

Loop executes for h value from 42 to 45.

After that, the else block is executed, and the loop is exited.

Q6

Respect

Solution

Initially, array holds p[20]= "Respect Your elders"

First while loop copies the array until the string encounters the null character of space. For given string after "Respect" loop terminates because of the space character.

In second while loop prints the "Respect" string which is stored in b array.

Q7

9 10 9 8

Solution

No Solution

Q8

It will print "welcome" only one time

Solution

No Solution

Q9

19

Solution

Initially arr[]={10,7,2,6,1,2,3}

For loop runs three time for p=0,1,2

arr[3] value added thrice (6+6+6 = 18)

Q10

50

Solution

Initially arr[]= {10, 20,30,40,5}

a= 20 +30 (1st and 2nd index value added and assigned to 'a')

As a result 50 gets printed.

Q11

2

Solution

In for loop b value varies from 0 to 1. (2 power 3 =8 , 1 power 8 =1)

For each iteration b increment by 1. So when b=1 (b=1+1)

So b gets printed.

Q12

0

Solution

Any number mod 1 gives answer 0.

Here is the math to illustrate how to get 20 mod 1 using our Modulo Method:

20 / 1 = 20

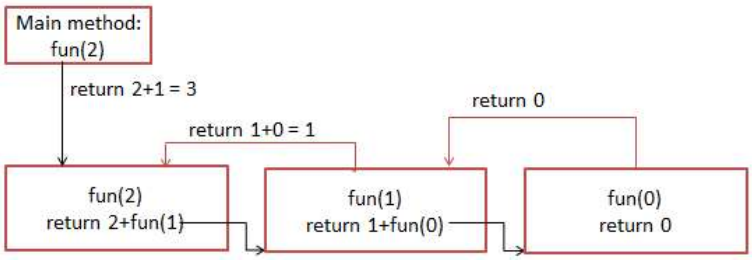
20 x 1 = 20

20 - 20 = 0

Thus, the answer to 20 mod 1 is 0.

Q13 3

Solution



Recursion starts from the main method.

Q14 10

Solution

Initially b=5, a=2, c=2

If block gets executed and assigns c value to a **(a=2)**

Second, if block also gets executed and increment c **(c=3)**

Sum of 2+5+3 = 10 gets printed.

Q15 10 12 15 86 777

Solution

No Solution

Q16 Open windows task manager and force closing the application

Solution

If a program has become completely unresponsive, you can press (and hold) Ctrl+Alt+Delete (the Control, Alt, and Delete keys) on your keyboard to open the Task Manager. You can then select the unresponsive application and click End task to close it.

Q17 Personalized marketing emails from product suppliers

Solution

Email marketing is the act of sending a commercial message, typically to a group of people, using email. In its broadest sense, every email sent to a potential or current customer could be considered email marketing. It involves using email to send advertisements,

request business, or solicit sales or donations.

Q18 Secondary storage devices

Solution

Popular secondary storage devices such as magnetic tape, magnetic disk, and optical disk

Q19 Voice over internet protocol (VoIP)

Solution

Voice over Internet Protocol (VoIP), is a technology that allows you to make voice calls using a broadband Internet connection instead of a regular (or analog) phone line.

Q20 Only 1

Solution

To send the email to a colleague who is not on the To or Cc lines.

Q21 Shift

Solution

To maintain the object's proportions, press and hold SHIFT while you drag the sizing handle.

Q22 Shortcut

Solution

The correct answer is Shortcut. The link to access any item on your computer or on a network such as a program, file folder, etc is known as Shortcut. In Microsoft Windows, a shortcut is a link that points to a computer program.

Q23 One-to-many

Solution

No Solution

Q24 Registers

Solution

The ALU of a computer normally contains registers for high-speed storage.

Q25 Space

Solution

Space cannot be the part of an email address.

Q26 Web downloading utilities

Solution

Web download utility that integrates with your Web browser to intercept any download operations that you initiate.

Q27 PNR number and date

Solution

A primary key, also called a primary keyword, is a key in a relational database that is unique for each record.

PNR number is a 10-digit number that you get against every train ticket you book and this is unique, which means the PNR number for each journey will be different so its the primary key.

Q28 Time Sharing

Solution

Explanation: The answer is time-sharing. In time-sharing, Users were able to interact with the computer and could share its information processing resources.

Q29 Only 3

Solution

The IFNA function in Excel returns an alternate value as specified by the user if a formula results in an #N/A error. The purpose of the IFNA function is to use an alternate value instead of the #N/A error.

Q30 Cookie stealing

Solution

Explanation: There are three major ways of stealing email information. These are by stealing cookies, social engineering and password phishing technique. The remaining three (in the option) are not ways of stealing email information.